



ClaimGuard AI

AI Health-Insurance Claims Adjudication & Fraud Defence

A Multi-Agent AI Decision Layer for Faster, Fairer & Fraud-Resistant Health Claims

Hackathon Prototype · Version 1.0.0
MVP Simulation — anonymized mock data only

THE PROBLEM, SOLVED WITH AI

Faster genuine claims. Defence against fraud. Transparency for every stakeholder in the health-insurance ecosystem.

~40,000

claims/day under PM-JAY

₹1.40L Cr

value processed annually

1,900+

treatment packages

100%

audit-trail coverage (target)

A working full-stack prototype: 14 UI pages · 21 REST endpoints · 10 verification agents · 35 automated tests.

Claims Fraud at National Scale

India processes a massive volume of health-insurance claims — manual review cannot keep up.

Each claim bundles bills, discharge summaries, diagnostic reports, treatment codes, patient identity and medical images. Most are genuine — but a meaningful fraction hide costly anomalies and fraud.



Context from the source PDF — shown for framing only; this prototype uses no live PM-JAY data.

EIGHT FRAUD & FAILURE TYPOLOGIES

- **Document manipulation** — Edited PDFs, altered watermarks, fake reports
- **Package mismatch** — Claim filed under a higher-value package
- **Duplicate procedures** — Same procedure claimed multiple times
- **Repeat admission** — Same patient readmitted for a similar issue
- **Reused images** — One diagnostic image across many claims
- **Inflated billing** — Consumables, room, implants over-billed
- **Ghost identity** — Fake or ineligible beneficiary
- **AI-generated forgery** — Fabricated clinical notes via AI tools

Why Manual Review Is Not Enough

The bottleneck is structural — not effort.



Delay

Every claim waits for a human to read it; genuine patients face discharge delays and pending reimbursements.



Inconsistency

Different auditors reach different judgments on similar claims — no uniform standard at scale.



Reactive

Fraud is typically detected after payment, when recovery is hard or impossible.



Cannot scale

Manual scrutiny simply cannot handle 40,000+ claims a day across 1,900+ packages.

MANUAL REVIEW TODAY

Slow · inconsistent · reactive · limited scale — fraud leakage and pendency grow with volume.

AI-ASSISTED REVIEW

Fast (<60s summary) · consistent rules · proactive (flagged before settlement) · population-scale.

ClaimGuard AI — The Decision Layer

An AI co-pilot between hospitals and payers: ingest → verify → score → explain → human-decide → log → learn.

SEVEN EXPLAINABLE DECISION OUTPUTS

Claim Score	0–100 trustworthiness
Fraud Risk	Low / Medium / High / Critical
Medical Necessity	Supported / Weak / Not Supported
Document Completeness	Complete / Missing / Suspicious
STG Compliance	Follows / Partial / Violates
Recommended Action	Approve / Query / Audit / Fraud hold
Explainability	Human-readable reasons + evidence

RISK BANDS DRIVE ROUTING

0–25	Low Risk	Auto-approve / Fast-track
26–50	Medium Risk	Query Hospital
51–75	High Risk	Medical Auditor Review
76–100	Critical Risk	Fraud Investigation Hold

WHAT MAKES IT UNIQUE

- Multi-agent — each agent owns one verification domain
- Human-in-the-loop — safe AI-assisted review, not blind automation
- Explainable — every score traces to named evidence
- NHCX-compatible by design; learns from auditor feedback

Beyond a Single Model

ClaimGuard combines specialized agents with statistics-based fraud forensics grounded in real audit findings.

10 specialized agents

Intake, OCR/Document, FHIR/NHCX, Identity, Coding, STG, Billing, Image, Fraud Pattern, Explainability — each a focused verifier.

Benford's-Law billing test

First-digit analysis with Nigrini MAD thresholds flags fabricated / manipulated billing amounts.

Ghost-beneficiary detector

Replicates real CAG findings — many beneficiaries on one phone, claims on deceased patients, double admissions.

Collusion network graph

Provider ↔ beneficiary ↔ broker links surface referral rings invisible in tabular data.

Provider peer-outlier radar

Z-score benchmarking flags hospitals deviating $>1.5\sigma$ from their peer cohort.

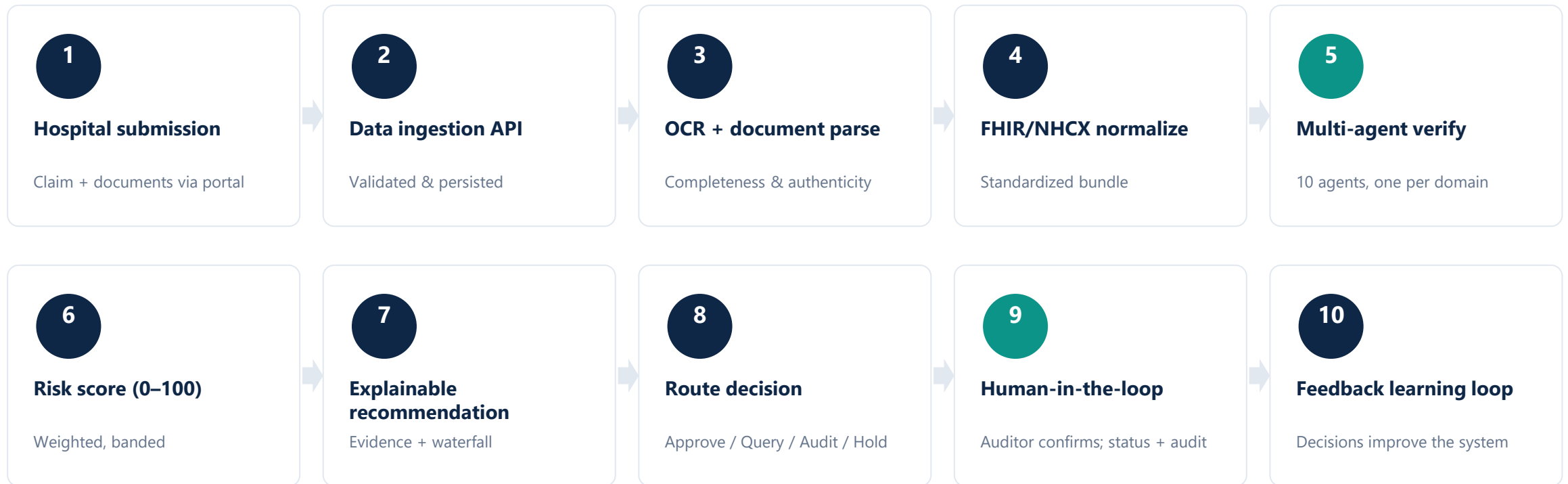
Exact score attribution

Because scoring is rule-based, the waterfall is exact — every point traceable, not estimated.

All techniques are real fraud-analytics methods applied to ANONYMIZED MOCK DATA — grounded in Nigrini, CAG PM-JAY audits, and graph-FWA literature.

From Submission to Settlement & Learning

A 10-step pipeline — fully implemented end-to-end in the prototype.



Steps 5–7 (verify → score → explain) and step 9 (human decision) are the heart of the system — each produces auditable, explainable output.

Three-Layer Architecture

Decoupled SPA + REST API + deterministic decision engines — buildable today, NHCX-ready tomorrow.

LAYER 1 · INTEGRATION

Hospital Portal · Insurer · PM-JAY · TPA · NHCX Gateway

Roadmap — represented, not live

LAYER 2 · AI DECISION

API Gateway · Claim Processing · Document Intelligence · Multi-Agent Orchestrator · Rules Engine · Vector KB → Risk Score + Explainable Recommendation

Orchestrator + rules engine implemented (deterministic); ML services placeholder

LAYER 3 · USER

Auditor Dashboard · Hospital Dashboard · Fraud Investigation · Policy Analytics

Implemented — all four dashboards live

SECURITY LAYER

- RBAC roles
- Audit trail
- Model-output logging
- Human override
- OAuth2 (roadmap)
- AES-256 (roadmap)
- Bias / FP monitoring
- Consent-aware data

Pure-function engines (O(signals), no model inference) → sub-millisecond per-claim latency; async FastAPI scales horizontally behind a gateway.

Stack & Deterministic Risk Engine

Proven, lightweight technologies; transparent, reproducible scoring.

TECHNOLOGY STACK

Frontend	React 18 · Vite · TypeScript · Tailwind · Recharts
Backend	FastAPI · Uvicorn · Pydantic v2
Persistence	SQLAlchemy 2 · SQLite (→ PostgreSQL)
Testing	Pytest · HTTPX TestClient — 35 tests
Packaging	Docker · docker-compose

RISK ENGINE — SCORE = min(Σ weights, 100)



Signals set by toggle OR derived from data (missing-doc, inflated-billing > rate×1.15). Fully auditable — no opaque model.

Built for Population Scale

Stateless engines and an async API path from a laptop demo to national volume.

Stateless engines

Risk, agent & forensic logic are pure functions over claim dicts — trivially parallelized and cached.

Async API

FastAPI + Uvicorn scales horizontally behind a gateway; no per-claim model inference in the MVP.

Production data path

SQLite → PostgreSQL + document store + dedicated audit service; vector KB for policy/STG lookups.

Optional ML services

LLM, fraud ML and image forensics plug in behind the same orchestrator interface.

MVP TARGET METRICS (from PDF)

Low-risk fast-track rate	50–70%
Manual-review reduction	40–60%
Document extraction accuracy	90%+
Fraud recall (high-risk)	80%+
False hold rate (genuine)	<5–8%
Claim summary generation	<60s
Auditor decision-time cut	50%+
Audit-trail coverage	100%

Aspirational project targets from the PDF — not measured prototype results.

Value for Every Stakeholder

One platform, six beneficiaries — speed for the genuine, defence against the fraudulent.

Govt Health Authority

PAIN

Fraud leakage & monitoring burden

CLAIMGUARD DELIVERS

Real-time fraud analytics & scorecards

Insurers / TPAs

PAIN

High review cost, inconsistency

CLAIMGUARD DELIVERS

Automated triage & consistent scoring

Claim Adjudicators

PAIN

Too many documents, unclear evidence

CLAIMGUARD DELIVERS

One-page AI summaries & reason-based flags

Medical Auditors

PAIN

Verifying medical necessity

CLAIMGUARD DELIVERS

STG-compliance support & evidence

Hospitals

PAIN

Delayed reimbursements

CLAIMGUARD DELIVERS

Faster payment for genuine claims

Patients

PAIN

Discharge delays & uncertainty

CLAIMGUARD DELIVERS

Faster approval, fewer disputes

Expected Impact

Even a 20–30% reduction in manual review at national scale means thousands of claims cleared faster every day.

60%

Faster settlement for low-risk claims

80%+

Fraud recall for high-risk claims

50%+

Auditor decision-time reduction

<60s

Claim summary generation

THE IMPACT TRIANGLE

Faster Claims

Speed for genuine claims

Lower Fraud

Defence against fraud

Higher Trust

Transparency for all

= Better Healthcare Access — funds stay available for genuine patients

PROOF OF BUILD

A Real, Working Full-Stack Prototype

Not slideware — a tested application you can run locally today.

14

UI pages / routes

21

REST API endpoints

10

Verification agents

12

Risk signals

12

Seed scenario claims

35

Automated tests — all passing

DELIVERED PAGES

Landing · Claims Dashboard · Claim Intake · Claim Review · Multi-Agent Runner · Decision Report · Analytics · Fraud Workspace · Forensics Lab · Hospital Portal · Audit Logs · Feedback Loop · Architecture · Demo Guide

Run it: `uvicorn app.main:app (backend :8000) + npm run dev (frontend :5173) → Vite proxies /api to the backend.`

From MVP to National Deployment

A staged path that hardens fraud ML, integrations and governance.

Phase 1

NHCX / FHIR-compatible integration

Phase 2

Advanced fraud ML + hospital-patient graph analytics

Phase 3

Image tampering & reuse detection (perceptual hashing, CNN)

Phase 4

STG compliance engine across more specialties

Phase 5

Real-time fraud heatmaps & hospital scorecards

Phase 6

Model governance, bias monitoring, audit compliance

Phase 7

Scalable deployment for state / national volume

Ongoing

Continuous learning from auditor feedback



Higher trust for everyone.

ClaimGuard AI brings speed for genuine claims, defence against fraudulent claims, and transparency for every stakeholder in the health-insurance ecosystem.

Innovation

Technical Feasibility

Scalability

User Impact

Working Prototype

MVP Simulation — anonymized mock data · no real PM-JAY / NHCX integration · no real patient data · every decision stays human-in-the-loop.